

G-04-geometric-depletion v1: audit geometric control of W_- bottleneck

ZFL Lab

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MISSION: Audit whether geometric depletion mechanisms provide quantitative control relevant to W_- bottleneck from E-strong-neg-02. INPUT E-strong-neg-02: W_- temporal control = OPEN/BLOCKED, $W_+ \in L^1_{loc}$ insufficient, $\int W_+ = \infty$ PROVEN from C_E^{weak} , C_E^{strong} requires $W_- \in L^1(0, T_*)$.

EXTERNAL DEPENDENCIES TO IDENTIFY: Constantin-Fefferman (CF93), Deng-Hou-Yu (DHY05/06). Exact theorem version to be verified.

FORBIDDEN: geometric alignment $\Rightarrow W_- \in L^1_t$ without explicit time estimate. citation $\Rightarrow$ theorem applicability. external criterion $\Rightarrow$ internal PROVEN status.

AUDIT:

A. Constantin-Fefferman family:

Exact hypotheses (CF93): In region of large $|\omega|$, direction $\xi = \omega/|\omega|$ Lipschitz: $|\sin \phi(x, y)| \leq |x - y|/\rho$ with $\rho > 0$, plus $|\omega| \in L^\infty$ control? Regularity of ξ in high vorticity region.

Exact geometric quantity: $\rho^{-1} =$ Lipschitz constant of direction, or $|\nabla \xi|$ in L^∞ .

Exact conclusion: If $\|\nabla \xi\|_{L^\infty}$ bounded and velocity bounded in suitable, then no blow-up? Actually CF gives $\|\omega(t)\|_\infty$ controlled exponentially by $\|u\|_\infty$ times geometric factor.

Whether controls W, W_- or other: Controls $D_t|\omega| = \alpha|\omega|$, $\alpha = \xi^T S \xi$. Note $W = \int \alpha|\omega|^2$. So geometric regularity controls α pointwise, not directly integral sign split W_- .

Scaling: $\nabla \xi \sim length^{-1}$, critical if $\rho \sim \sqrt{T_* - t}$.

Time-integrability: CF requires $\int_0^{T_*} \|u\|_\infty^2 dt$? No L^1_t for W_- directly. Estimate: $|\alpha| \leq C\|u\|_{Lip}$ but not sign.

Applies near T_* ? Requires uniform Lipschitz of ξ in high vorticity region up to T_* , strong assumption.

Bridge to C_E^{strong} ? No direct $W_- \in L^1$. Only gives bound on $|\alpha|$, i.e., $|W| \leq C \int |\omega|^2 \|u\|_{Lip}$.

B. Deng-Hou-Yu family:

Exact hypotheses (DHY): Conditions on vortex line length $L(t)$, curvature κ , divergence of ξ , plus U_ξ, U_n velocities.

Exact geometric quantity: $L(t)$ of vortex line segment, $\int_0^L (\nabla \cdot \xi)_+$, $M(t) = \max |\omega|$ along line, etc.

Exact conclusion: If $L(t) \geq cL_0$, curvature and divergence controlled in L^∞ , and U_ξ, U_n bounded, then no blow-up up to T_* .

Controls W, W_- or other: Controls growth of $\|\omega\|_\infty$ via Lagrangian stretching of vortex lines: $\partial_t |\omega|/|\omega| \sim$ divergence + curvature terms. Again controls $|\alpha|$, not sign split.

Scaling: Length $L(t)$ critical if $\sim (T_* - t)^{1/2}$? Actually geometric non-collapsing.

Time-integrability: Conditions are pointwise in time L^∞_t bounds on geometric quantities, not L^1_t for W_- .

Applies near T_* ? Yes, designed as non-blow-up criterion up to T_* .

Bridge to C_E^{strong} ? Would need to extract $W_- = \int (-\alpha)_+ |\omega|^2$. If α bounded above by geometric quantity $G(t)$, then $W_- \leq \int G_- |\omega|^2$? But G not sign definite. Still no L^1 .

OUTPUT:

CF: controls $|\alpha|$, not $W_- \in L_t^1$ directly

DHY: controls $\|\omega\|_\infty$ growth, not W_- integrability

geometric alignment $\not\Rightarrow W_- \in L^1(0, T_*)$ without extra temporal bridge

BRIDGE to C_E^{strong} = OPEN/BLOCKED: PARTIAL BRIDGE only on $|\alpha|$

Hence E-strong-neg-02 remains bottleneck: geometric depletion controls magnitude of stretching, not its signed decomposition required for $\int W_+ = \infty \Rightarrow C_E^{strong}$.

Potential new pathway: geometric quantity $G(t) := \|\nabla \xi\|_\infty$ or (L, κ, U) as independent C_G^{geom} candidate, to be crossed with C_{L^3} and C_G^{CKN} , rather than forcing W_- control.

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ untouched.

Next: CROSS-Ggeom-L3-CKN if G exact version identified.